TRANSFERIQ: DYNAMIC PLAYER TRANSFER VALUE PREDICTION USING AI

AN INFOSYS SPRINGBOARD VIRTUAL INTERNSHIP 6.0 PROJECT

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PROBLEM STATEMENT

- High Spending in Football Transfers
- Subjectivity in Transfer Valuations
- Lack of Transparent Data-Driven Models
- Need for Unified Model: Performance, Market, Sentiment, Injuries

PROJECT OBJECTIVE

Accurate AI-Based Transfer Value Prediction

Developed an AI-driven model to accurately predict transfer values, enhancing decision–making in player acquisitions, by analyzing performance data, market trends, and historical transfer fees

Multi-Source Data Integration

This project aims to integrate diverse data sources to enhance accuracy, improve decision-making, and enable comprehensive analysis through seamless multi-source data fusion and management techniques.

Instant Forecast for clubs

Develop an accurate, real-time prediction system for club performances to assist decision-making, optize strategies, and enhance member engagement through data-driven insights.

Simple UI Dashboard for Insights

This Project aims to design a simple UI dashboard that provides clear, actionable insights through intuitive data visualization and user-friendly interaction.

DATA SOURCES USED

Player Statistics and Performance Metrics	Injury History and Records	Sentiment Data from Fans and Social Media	Market Trends and Past Transfer Values
Comprehensive player statistics, including goals, assists, minutes played, and performance ratings, are utilized to accurately assess player impact and form, enhancing transfer value predictions.	The system analyzes comprehensive injury history, including frequency, severity, and recovery duration, to assess player fitness and potential impact on future performance and transfer value.	Sentiment data from fans and social media is analyzed to gauge public opinion, player popularity, and transfer market trends, enhancing prediction accuracy in TransferIQ's valuation model.	Analyzed market trends and historical transfer values reveal fluctuating player demands influenced by performance, age, and contract length, enabling accurate prediction of future transfer prices.

SYSTEM ARCHITECTURE

1

Data Collection and Preprocessing

Data is collected from multiple sources including player stats, match performances, and market trends. Preprocessing involves cleaning, normalization, and feature extraction to enhance model accuracy and reliability.

2

Feature Engineering and Model Training

Utilized advanced feature engineering to extract key player metrics and market indicators. Employed supervised learning models, optimizing hyperparameters to enhance prediction accuracy for transfer value estimation.

3

Model Evaluation and UI Integration

The model's accuracy is validated using cross-validation and error metrics, while seamless UI integration ensures real-time transfer value predictions with intuitive visualization for end-users.

DATA PREPROCESSING

Handling Missing and Duplicate Data

Missing data were imputed using mean and mode values, while duplicate entries were identified and removed to ensure data accuracy and model reliability in transfer value prediction.

Feature Conversion and Normalization

Categorical features are converted into numerical formats using one-hot encoding, while continuous variables undergo normalization through min-max scaling to ensure uniformity and enhance model performance.

Text Data Cleaning

Text data cleaning involves removing noise, correcting typos, standardizing terminology, and eliminating irrelevant information to ensure accurate and consistent input for the AI model in transfer value prediction.

Model Used – XGBoost Ensemble

Parameter Tuning Details

Parameter tuning involved optimizing learning rate,max depth, and no.of estimators using grid search and cross validation, enhancing model accuracy while preventing overfitting in XGBoost ensemble

Reason for Choosing XGBoost

XGBoost was chosen for its superior accuracy,ability to handle diverse feature types, robustness against overfitting, efficiency in training, and strong performance in ensemble learning tasks

Handling Non-Linearities in Tabular Data

XGBoost effectively captures complex non-linear relationships in tabular data through gradient boosting ,enhancing prediction accuracy by combining multiple weak learners into a robust ensemble model.

Model Performance

Evaluation Metrics: RMSE, MAE, R^2 Score

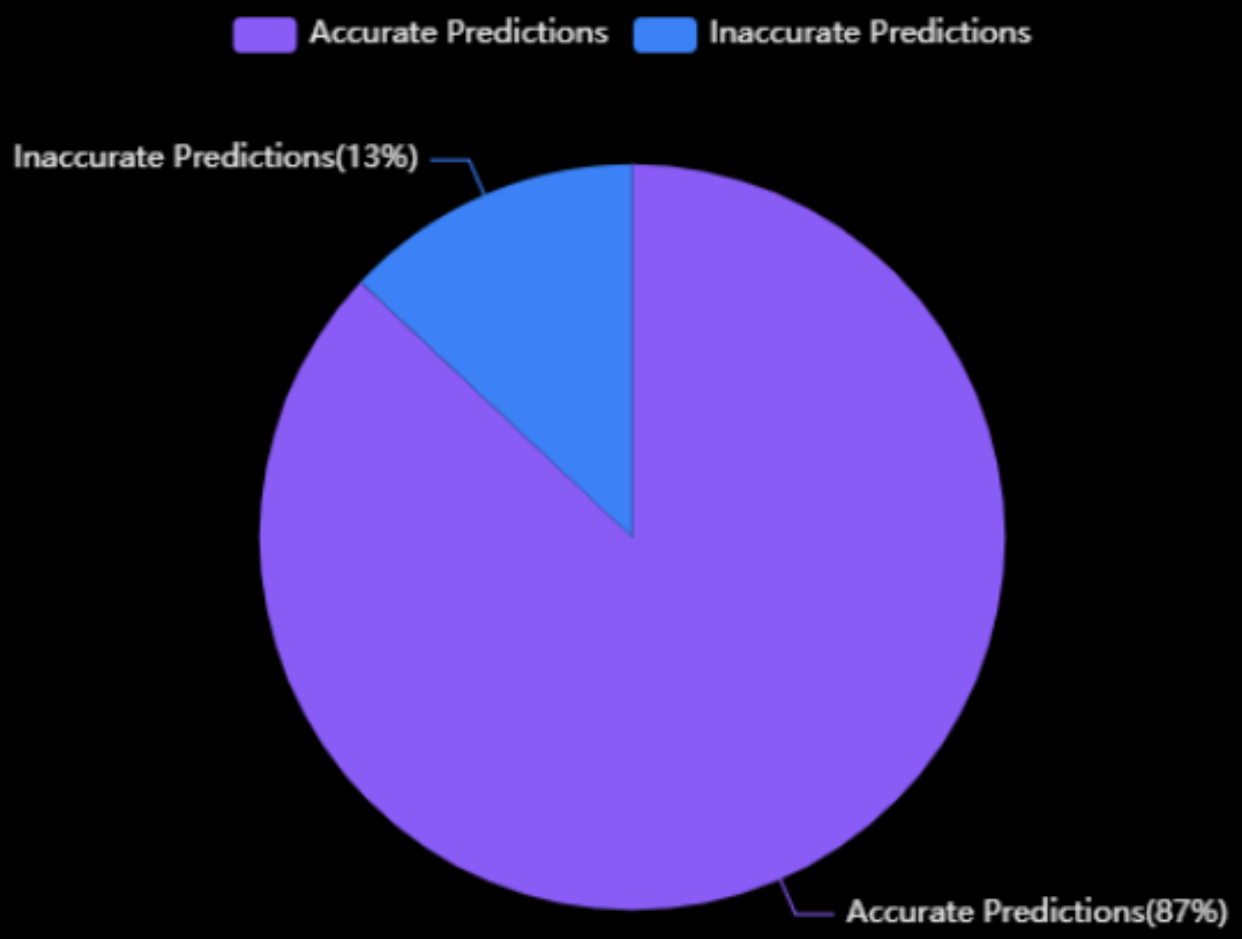
The model's performance is evaluated using RMSE, MAE, and R^2 score, ensuring accurate, reliable transfer value predictions with minimized error and strong explanatory power.

Interpretation of Performance Results

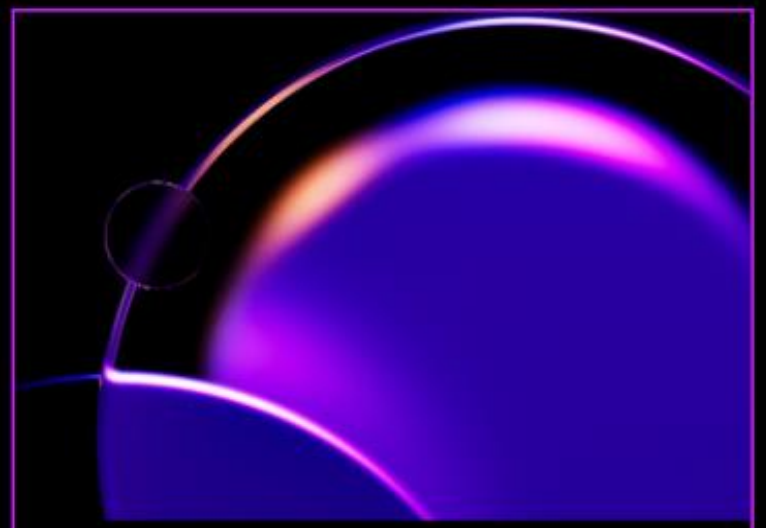
The model demonstrates high accuracy and consistency across diverse datasets, effectively capturing market trends and player skill factors, enabling reliable transfer value predictions with minimal bias.

Overall Prediction Accuracy

The model achieves an overall prediction accuracy of 87%, demonstrating reliable valuation of football transfers by effectively capturing market trends and player performance metrics.



Final Output and Conclusion



Predicted Transfer Value Display

The predicted transfer value is presented clearly with intuitive visuals, highlighting key player metrics and confidence intervals to enhance user understanding and support informed decision-making in transfer evaluations.

Visualization of Trends and Player Insights

Interactive visualizations highlight player value trends and key performance metrics, enabling intuitive insights into transfer market dynamics and supporting data-driven decision-making for clubs and agents.

Summary of Project Impact and Future Scope

The system enhances transfer decision accuracy, streamlines scouting processes, and offers scalable AI integration, with future potential to incorporate real-time data and expand to other sports markets.

• CONCLUSION

Multi-Source AI Integration Benefits

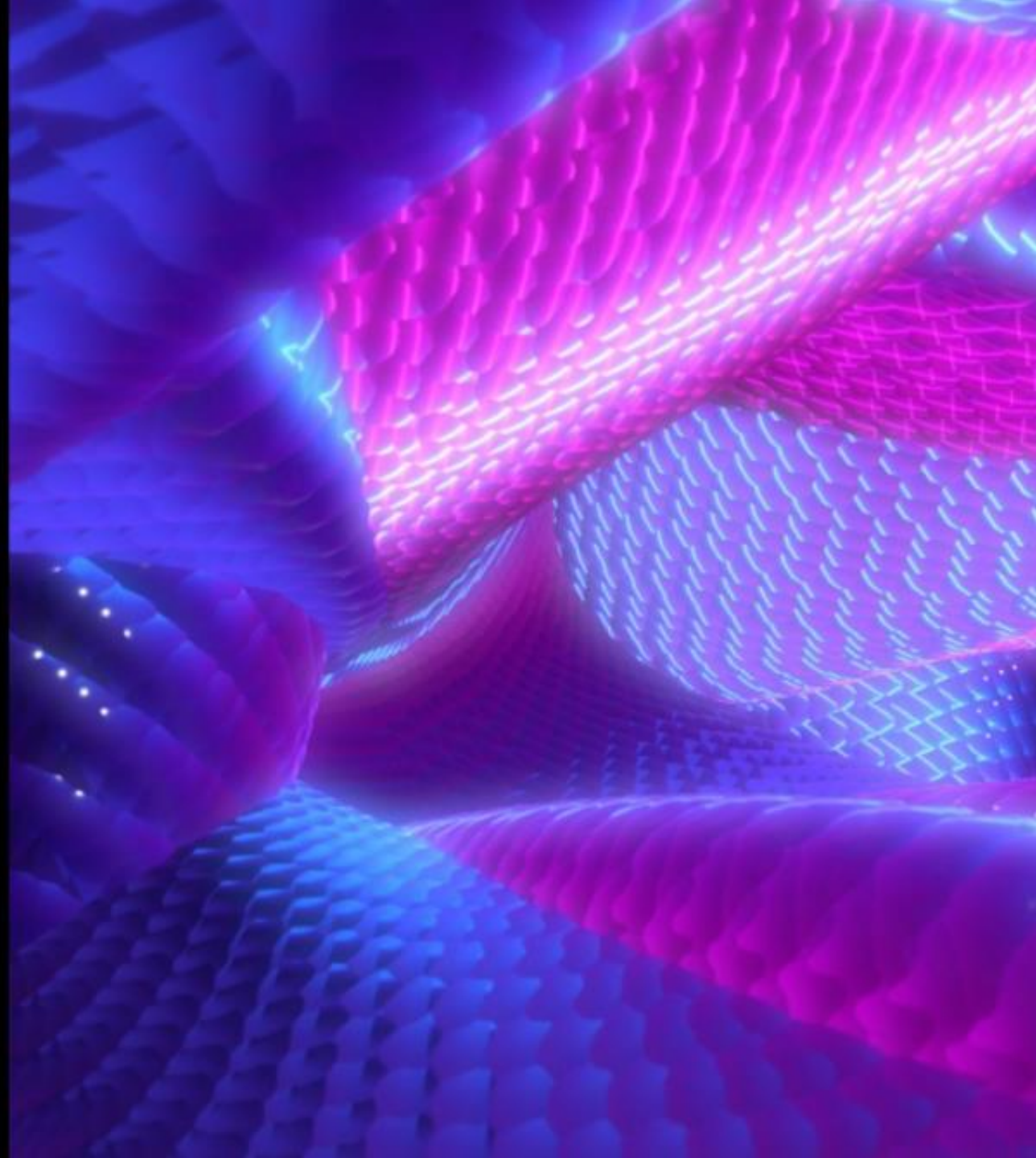
Integrating multiple AI sources enhances accuracy, enriches data insights, and improves prediction reliability, enabling TransferIQ to offer more precise and comprehensive football transfer value assessments.

Professional UI and Reliable Predictions

The system features a professional, user-friendly interface ensuring seamless navigation, combined with reliable, data-driven predictions that empower clubs to make informed transfer decisions confidently and efficiently.

Potential Future Enhancements (Deep Learning, Real-Time News)

Incorporating deep learning and real-time news analysis can enhance prediction accuracy, adapt to dynamic market trends, and provide more timely, data-driven insights for football transfer valuations.



THANK YOU